



expected to accelerate the growth of the AOI market in the years to come. Nordson offers diverse AOI solutions for the electronics and semiconductor sectors. Their systems are designed to deliver accurate and reliable inspection results, helping manufacturers comply with rigorous quality standards.

As a major player in the AOI market, KLA provides advanced inspection solutions for the semiconductor industry. Their AOI systems are crucial for detecting defects in wafers and other semiconductor components, ensuring high precision and quality.

CyberOptics is a leading provider of AOI solutions recognised for their high-speed inspection capabilities and exceptional accuracy in defect detection. Their systems play a key role in the electronics manufacturing industry.

ViTrox is a key player in the AOI market, offering various inspection solutions tailored for the electronics and semiconductor industries. Their AOI systems are designed to meet the stringent quality control standards essential for these sectors.

Renowned for its innovative AOI solutions, Saki Corporation combines high-speed inspection with high-resolution imaging to deliver superior performance. Their AOI systems are widely adopted in the electronics manufacturing to ensure high product quality and reliability.

Some key players

NMTronics offers a comprehensive range of AOI systems from renowned manufacturers such as Mirtec, Saki, and Panasonic. These systems leverage cutting-edge optical and AI technologies to enhance the inspection of solder joints, component placements, and other critical aspects of PCBs. Here is a closer look at the capabilities

and features of these advanced systems:

Mirtec AOI systems. Mirtec's MV-7 series exemplifies vision-based inspection, using high-speed 3D technology combined with AI to detect defects accurately. Another standout model, the MV-6, is designed for inline integration, ensuring real-time defect detection during manufacturing. For soldering inspections, the MV-3 focuses on real-time analysis of solder joints, effectively supporting wave and reflow soldering processes.

Panasonic AOI systems. Panasonic's NPM-W2 and NM-X series have high-definition cameras and AI-driven image processing, delivering precise and fast inspection results. Notably, the NPM-W2 incorporates intelligent vision and machine learning algorithms, enabling it to adapt seamlessly to changes in PCB designs and manufacturing processes. This results in superior defect detection accuracy and improved throughput. Moreover, its 3D inspection capabilities allow for identifying visible and hidden defects by scanning PCBs in multiple dimensions.

Saki AOI systems. Saki is a leader in 3D AOI technology, with multi-angle imaging systems to inspect solder joints and components from all perspectives. This approach ensures a more comprehensive inspection compared to traditional 2D methods. Saki also offers advanced 3D AOI systems integrated with X-ray technology, providing a detailed analysis of complex PCB components and hidden features.

A call to action for electronics manufacturers

If you are in electronics manufacturing, AOI is not just a tool—it is your competitive edge. It ensures your products leave the production line flawless, protects your brand

reputation, and reduces costs in the long run. In an industry where even a fraction of a millimetre matters, AOI is the precision eye that ensures your PCBA is perfect every time. The next time you pick up a smartphone or switch on a smart appliance, remember: AOI is the silent hero, making it all possible.

Imagine stepping into a smart factory where engineers use augmented reality (AR) glasses to see inspection data directly on live PCBs. These glasses, paired with cloud analytics, can highlight defects, show real-time yield metrics, and suggest fixes. With electronics manufacturing growing more complex, inspection technologies like AOI, X-ray, and hybrids are key to ensuring quality and efficiency.

Emerging tools like AI, 3D imaging, and quantum dot sensors are reshaping automated inspection. Coming trends include quantum AOI for ultra-high-resolution imaging, cobots for dynamic inspections, and AI-driven predictive analytics for proactive defect control. The inspection field is evolving fast, with breakthroughs like quantum imaging enabling atomic-scale defect detection, autonomous systems reducing human input, and AR wearables improving defect visualisation. These technologies make inspections more precise and efficient, which is crucial as devices become more complex.

Optical, AOI, laser, and X-ray systems remain critical in electronics manufacturing. With AI, edge computing, and hybrid systems, inspection technology will continue advancing to meet modern challenges, ensuring reliable, high-performing devices. **EFY**

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